

Lecture 22: Testing of Hypotheses (Introduction)

Start with a given parametric probability model with one or more unknown parameters.

For some r.v. X following this prob. distn f_θ , we have

$X \sim f_\theta$, $\theta \in \Theta$, the parameter space, and

$\mathcal{P} = \{f_\theta, \theta \in \Theta\}$, family of distributions of which f_θ is a member.

A hypothesis is a conjecture or assertion about f_θ

Null hypothesis (H_0): a hypothesis that is believed to be true

Alternative hypothesis (H_A): an alternative belief.

Notationwise, $H_0: X \sim f_\theta \in \mathcal{P}_0 \subset \mathcal{P} \Leftrightarrow H_0: X \sim f_\theta, \theta \in \Theta_0 \subset \Theta$, or, simply
 $H_0: \theta \in \Theta_0$ (given $X \sim f_\theta, \theta \in \Theta$), $\mathcal{P}_0 = \{f_\theta: \theta \in \Theta_0 \subset \Theta\}$

and $H_A: X \sim f_\theta \in \mathcal{P}_A \subset \mathcal{P} \Leftrightarrow H_A: X \sim f_\theta, \theta \in \Theta_A \subset \Theta$, or,
 $H_A: \theta \in \Theta_A$ (given $X \sim f_\theta, \theta \in \Theta$), $\mathcal{P}_A = \{f_\theta: \theta \in \Theta_A \subset \Theta\}$.

Clearly, $\mathcal{P}_0 \cap \mathcal{P}_A = \emptyset$ (or, $\Theta_0 \cap \Theta_A = \emptyset$)

H_A is a restricted alternative if $\mathcal{P}_0 \cup \mathcal{P}_A \subset \mathcal{P}$ (or, $\Theta_0 \cup \Theta_A \subset \Theta$)
unrestricted if $\mathcal{P}_0 \cup \mathcal{P}_A = \mathcal{P}$ (or, $\Theta_0 \cup \Theta_A = \Theta$).

e.g., $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$
 $H_0: \theta = \theta_0$ (fixed) against $H_A: \theta > \theta_0$ (restricted H_A)
 $H_A: \theta \neq \theta_0$ (unrestricted H_A)

Simple ~~Null~~ (or, Alternative) Hypothesis

When the distribution f_θ is completely specified by H_0 (or, H_A).

e.g., $X \sim N(\theta, 1)$, $H_0: \theta = \theta_0$ (fixed) ag. $H_A: \theta = \theta_1$ (fixed).

Composite ~~Null~~ (or, Alternative) Hypothesis

When the distribution f_θ is only partially specified by H_0 (or, H_A)

e.g., $X \sim N(\theta, 1)$, $H_0: \theta \leq \theta_0$ ag. $H_A: \theta > \theta_0$.

All 4 combinations

	Simple	Composite
H_0	.	.
H_A	.	.

are possible.

In general, the problem is to design a test for testing H_0 ag. H_A .

This is done by constructing a rejection (of H_0) or critical region, say, $C \subset X$, the sample space, and hence, a partition of X , $\{C, X-C\}$

If the observed $x \in C$, we reject H_0
 $x \in X-C$, 'accept' H_0 (ag. an unrestricted H_A) | WHY?
'do not reject' H_0 (ag. a restricted H_A)

Question How to construct C , i.e., create such a partition of X ?
What should be the criteria for construction of such a partition?

While creating C (and hence, $X-C$) we commit 2 types errors

		True Situation	
		H_0 true	H_0 false / H_A true
Decision Taken	Accept / Do not rej. H_0	✓	Type II error
	Reject H_0	Type I error	✓

Errors are random events since they involve random observations, viz,
Type I Error: $x \in C$ when H_0 true
Type II Error: $x \in X-C$ H_A

So, we consider probs. of such events.

Probability of Type I error = $P(x \in C | H_0) = P_0(x \in C | \theta \in \Theta_0)$; and
Probability of Type II error = $P(x \in X-C | H_A) = P_0(x \in X-C | \theta \in \Theta_A) = 1 - P_0(x \in C | \theta \in \Theta_A)$

Ideally, we should construct the critical C in such a way so that both the probabilities are minimized. But that is not possible, increasing one, leads to the decrease in the other. WHY?

Next, some more definitions,
the principle of constructing C &
an example